

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering ASSESSMENT TOOL 1 – SHORT ANSWER TEST

Course Code:	CSA16	Course Title:	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL1, BL2)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	10% of CIA	Total Marks:	100 (5 Questions × 20 Marks)
CO1 Statement:	Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2		
Student Name:	N. Sunil Kumar	Reg. No.:	192472330
Section / Batch:	CSE(A1)	Date:	

Instructions to Students

- Answer all questions. Each question carries 20 Marks. Total: 100 Marks.
- Clearly show all requirements, process, operations.
- Use appropriate models, schema represented scenario wise
- Justify each step in different dimension specified and measures clearly.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Short Answer Test	Architecture of Data Warehouse, ETL, OLAP	Q1 – Q5	Q1 –20 Q2 –20 Q3 –20 Q4 –20 Q5 –20
GRAND TOTAL:			100 Marks

SECTION A – SHORT ANSWER TEST

1. Suppose that a data warehouse for City Hospital consists of the following four dimensions: patient, doctor, department, and time, and two measures visit_count and avg_charge. At the lowest conceptual level, avg_charge stores the actual consultation fee charged for a patient visit. At higher conceptual levels, it stores the average fee.

(a) Draw a snowflake schema diagram for the data warehouse.

(b) Starting with the base cuboid [patient, doctor, department, time], what specific OLAP operations should be performed to list the average consultation fee department-wise for each year?

(c) If each dimension has four levels (including all), how many cuboids will this cube contain (including base and apex cuboids)?

2. A smart city project is collecting data from multiple sources, including:
- Traffic sensors for vehicle counts and congestion levels.
 - IoT devices for monitoring air quality, noise levels, and energy consumption.
 - Public transportation systems for ridership data and punctuality.
 - Citizen feedback from a mobile app.

Questions:

Fact and Dimension Design:

- Design a star schema to analyze traffic congestion patterns, energy consumption trends, and public transportation efficiency.
- What measures and dimensions would you include? Justify your design.

3. Suppose that a banking data warehouse contains two fact tables: Transaction_Fact and Loan_Fact, sharing dimensions customer, branch, time, and account_type.

- Transaction_Fact → transaction_count, amount
- Loan_Fact → loan_amount, interest

(a) Draw a galaxy schema diagram.

(b) Identify OLAP operations to obtain branch-wise average transaction amount and loan amount yearly.

(c) If each dimension has four levels, calculate total cuboids.

4. A manufacturing company monitors product quality and production efficiency. The data warehouse contains:

- Dimensions:
 - Product: Category, type, and batch.
 - Factory: Location, department, and line number.
 - Time: Year, quarter, month, and week.
 - Supplier: Region, rating, and contract type.
- Measures:
 - Defect rate.
 - Production units.
 - Supplier reliability percentage.

Questions:

- Roll-up: Summarize defect rates by product category for the current year across all factories.
- Drill-down: Focus on a specific factory to analyze defect rates by department and product type.
- Slice: Isolate production unit data for suppliers with a rating of 4+ across all regions.
- Dice: Analyze defect rates and supplier reliability for selected product categories and factory locations.

5. A retail chain manages inventory and sales data across multiple stores. The data warehouse contains:

- Dimensions:
 - Product: Category, subcategory, and SKU.
 - Store: Region, city, and store ID.
 - Time: Year, quarter, month, and day.
 - Customer: Membership tier, age group, and gender.
- Measures:
 - Inventory turnover.
 - Total sales revenue.
 - Units sold.

Questions:

1. Roll-up: Summarize inventory turnover by product category across all stores for the last quarter.
2. Drill-down: Focus on sales data for electronics in a specific region, analysing by subcategory and SKU.
3. Slice: Isolate sales revenue data for male customers in the Bronze membership tier across all stores.
4. Dice: Analyze sales revenue and units sold for selected cities and product categories during specific time periods.

Code of Conduct

I N. Sunil Kumar (19247233) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____



RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Marks Evaluation:

Question No.	Understanding of Concepts (25%)	Explanation (25%)	Application (20%)	Organization (15%)	Accuracy (10%)	Clarity (5%)	Total Marks
Q1 (20 Marks)							
Q2 (20 Marks)							
Q3 (20 Marks)							
Q4 (20 Marks)							
Q5 (20 Marks)							
Overall Total (100)							

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Assessment Tool - 01

CO1AT1

CSA-1613-Data warehouse &

Data mining.

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① Hospital data warehouse :-

①a Snowflake scheme :-

fact table :- visit-fact

Dimension :-

Measures :-

patient-ID

Visit-count

Doctor-ID

Avg-charge

Department-ID

Time-ID

Dimension Tables :-

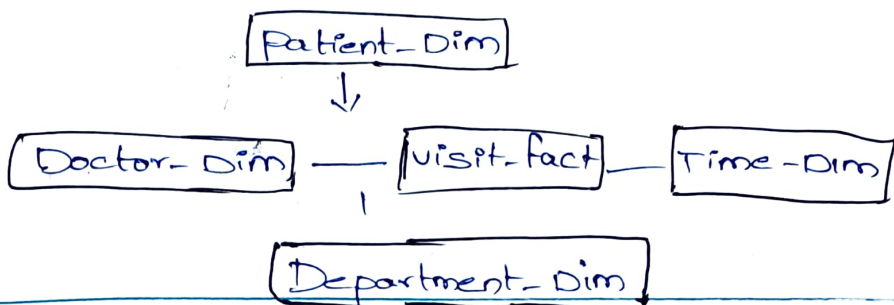
* patient-Dim → patient-ID, Name, Age, Gender

* Doctor-Dim → Doctor-ID, Name, Specialization

* Department-Dim → Department-ID, Dept-Name

* Time-Dim → Day, month, Quarter, year

Snowflake Structure :-



(b) OLAP operations :-

To find avg consultation fee department-wise for each year;

* Roll up :- Aggregate data from month $\rightarrow$ year

* Dice :- Select specific departments and years.

(c) Total cuboids :-

Dimensions = 4

Cuboid $\# = 2^4 = 16$ cuboids.

(2) Smart city Data warehouse :-

Measures :-

* vehicle-count

* Air-Quality index.

* Congestion-level

* punctuality.

Dimensions :-

1. Time Dimension :-

* Day, month, year

2. Location Dimension :-

* city, Area, Zone.

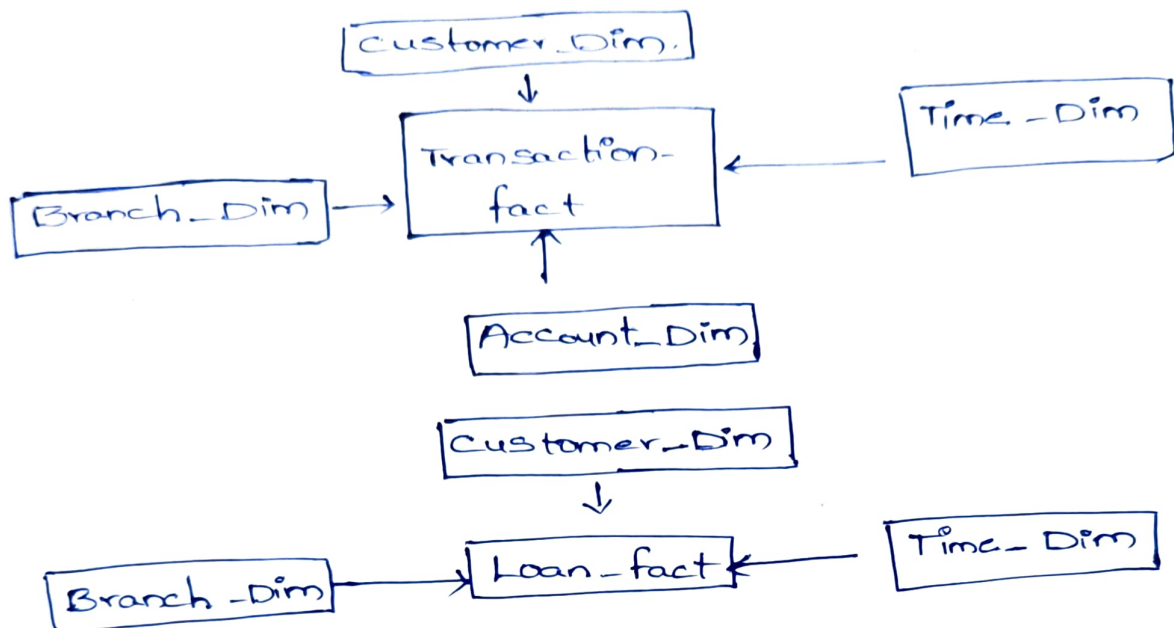
3. Transport Dimension.

* vehicle type Route.

* Justification :- This design helps to analyze traffic, pollution, energy usage & transport efficiency.

3) Banking Data warehouse :-

a) Star Schema :-



b) OLAP operations :-

for branch-wise Avg transaction & loan amount.

- * Roll up :- Branch - Region
- * Slice :- Select a particular year.
- * Dice :- Select branches & Loan types.
- * pivot :- Compare branches with measures.

c) Total cuboids :-

Dimensions = 4

cuboids = $2^4 = 16$ cuboids.



4) Manufacturing Data warehouse :-

Roll up :-

- * Defect rates by product category for curr y.

operations:-

- * product $\rightarrow$ category.
- * Time $\rightarrow$ year.

Result:- Avg defect rate for each product category across factories.

Drill down:-

factory $\rightarrow$ Department $\rightarrow$ product type.

Example:- factory A $\rightarrow$ production Department $\rightarrow$ electronics products.

Slice:-

Select:-

- * Supplier rating = 4+
- * All regions.

Dice:-

Analyze:-

- * Selected product categories
- * Selected factory locations.

⑤ Retail chain Data warehouse:-

Roll up:-

Inventory turnover by product category.

Hierarchy:- product $\rightarrow$ category.

Result:- Total inventory turnover for each category across stores.

Drill down:-

Analyze electronic sales:-

Region $\rightarrow$ Store $\rightarrow$ Sub category $\rightarrow$ SKU.

Slice:-

Conditions:-

- * Gender = Male
- * membership = Bronze

Dice:-

Select:-

- * Specific cities
- * time periods.